**DRIVER DIZZINESS DETECTION**

**Project Documentation and Evaluation Report**

**Project Title:** Driver Dizziness Detection
**Domain:** Artificial Intelligence, Machine Learning and Computer Vision
**Programming Language:** Python
**Libraries:** OpenCV, MediaPipe, NumPy
**Project Type:** Real-Time Computer Vision Application

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## 1. Abstract

Driver Dizziness Detection is a computer-vision-based monitoring prototype designed to identify visual indicators associated with driver drowsiness, particularly prolonged eye closure and possible yawning. The system uses a webcam to capture facial images and MediaPipe Face Mesh to detect facial landmarks. OpenCV processes these landmarks to estimate eye and mouth openness.

When eye closure or mouth opening continues beyond a configured duration, the system displays a warning and activates an audible alarm. It also handles moderate head movements, displays real-time measurements, and records detection events in CSV files for evaluation.

The prototype has been tested under different lighting conditions, with glasses, and during head movements. Its evaluation scripts report classification metrics and event-detection timing. Further testing with larger datasets and independently verified observations is required before making claims about real-world reliability.

## 2. Introduction

Driver fatigue and drowsiness can reduce concentration, awareness, and reaction time while driving. Computer vision offers a way to monitor visible facial indicators using a camera without requiring wearable sensors.

This project implements a real-time monitoring system that tracks facial landmarks and analyzes eye closure and mouth opening. It provides visual and audible alerts when configured conditions are met.

The project demonstrates the practical application of computer vision, real-time image processing, facial landmark detection, event tracking, and performance evaluation.

## 3. Problem Statement

There is a need for an accessible monitoring prototype that can identify selected visual signs of possible driver drowsiness and alert the user when those signs persist.

The proposed system uses a webcam and facial landmark measurements to monitor eye closure and mouth opening. It aims to provide timely warnings while maintaining detection during moderate head movements.

## 4. Objectives

- Detect a face in a live webcam feed.
- Track facial landmarks using MediaPipe Face Mesh.

- Calculate the Eye Aspect Ratio (EAR) to estimate eye openness.
- Calculate the Mouth Aspect Ratio (MAR) to estimate mouth opening.
- Detect prolonged eye closure and possible yawning.
- Activate an audible alarm when configured warning conditions are reached.
- Handle moderate head tilts and left/right face turns.
- Record measurements and event timings in CSV files.
- Evaluate classification performance and detection delays.

**5. Tools and Technologies**

**Python:** Main programming language used to implement the application.

**OpenCV:** Used for webcam access, frame processing, displaying measurements, drawing facial landmarks, and estimating head pose.

**MediaPipe Face Mesh:** Used to detect facial landmarks around the eyes, mouth, and other facial regions.

**NumPy:** Used for numerical calculations involved in landmark measurements and geometric operations.

**Threading:** Used to support background alarm operations.

**Winsound:** Used to generate audible warning tones on Windows.

**CSV:** Used to store measurement logs, event timings, and evaluation results.

**6. System Architecture**

The system follows this processing sequence:

1. Capture a frame from the webcam.
2. Detect facial landmarks.
3. Calculate eye and mouth measurements.
4. Apply smoothing and head-pose reliability checks.
5. Track the duration of eye closure and mouth opening.
6. Display the current status and measurements.
7. Activate the alarm when a configured warning condition is reached.
8. Record measurement and timing data.
9. Stop the application when the user presses Q.

**7. Detection Methodology**

**7.1 Facial Landmark Detection**

MediaPipe Face Mesh identifies facial landmarks from the webcam feed. Selected landmarks around the eyes and mouth are used to calculate geometric measurements.

### 7.2 Eye Aspect Ratio

The Eye Aspect Ratio (EAR) estimates eye openness by comparing vertical eye distances with horizontal eye width.

A lower EAR generally indicates that the eye is more closed. The system monitors the eye measurement over time rather than treating every brief blink as a prolonged-closure event.

**Configured parameters:**

- EAR threshold: 0.20
- Prolonged eye-closure duration: 1.5 seconds

When the configured eye-closure condition persists for the required duration, the system displays a warning and activates the alarm.

### 7.3 Mouth Aspect Ratio

The Mouth Aspect Ratio (MAR) estimates mouth opening using distances between selected mouth landmarks.

**Configured parameters:**

- MAR threshold: 0.35
- Possible-yawn duration: 1.0 second

When mouth opening persists beyond the configured duration, the system displays a possible-yawn warning and increments the yawn counter once for that continuous event.

Mouth opening is treated as an indicator of a possible yawn, not definitive proof that a yawn has occurred.

### 7.4 Smoothing

The system uses a smoothing factor of 0.4 to reduce rapid fluctuations in facial measurements. Smoothing helps stabilize threshold-based decisions, although it may also affect response timing.

### 7.5 Head-Pose Handling

The system estimates head pose using facial landmarks and OpenCV's solvePnP function. It applies a correction strategy for moderate yaw and avoids relying on measurements when the face is turned too far for reliable detection.

Testing reported for left/right head tilts and left/right face turns passed. More extensive testing is still necessary to establish performance across different users and camera positions.

### 7.6 Audible Alarm

The application uses a background thread to manage the alarm. The warning sound is triggered when the configured prolonged eye-closure condition is reached. The system is designed to stop the warning when the relevant condition clears.

## 8. Data Logging

The project uses CSV files to support evaluation and debugging.

| File | Purpose |
|---|---|
| detection_log.csv | Stores regular measurements, status, and possible-yawn count |
| event_timing_log.csv | Stores event type, event start, warning time, and detection delay |
| real_eye_tests.csv | Contains manually recorded eye-closure test cases |
| real_yawn_tests.csv | Contains manually recorded possible-yawn test cases |
| evaluation_report.csv | Stores classification evaluation results |
| complete_evaluation_report.csv | Stores combined classification and timing results |

The timing log records timestamps relative to the start of the camera session. Detection delay is calculated from the software-recorded event start to the warning time.

This timing calculation does not independently establish the exact physical onset of eye closure or yawning.

**9. Evaluation Method**

The project uses evaluate_complete.py to calculate classification metrics and summarize event-timing measurements.

The reported classification evaluation contains five cases for eye closure and five cases for possible yawning. Each category includes two positive cases and three negative cases.

The metrics include:

- **Accuracy:** Proportion of all evaluated cases classified correctly.
- **Precision:** Proportion of predicted positive cases that were true positives.
- **Recall:** Proportion of actual positive cases detected.
- **F1-score:** Harmonic mean of precision and recall.
- **False alarm rate:** Proportion of negative cases incorrectly classified as positive.

**10. Classification Results**

The following results were reported by the current evaluation script.

| Metric | Eye Closure | Possible Yawning |
|---|---|---|
| Tests evaluated | 5 | 5 |
| True positives | 2 | 2 |
| False positives | 0 | 0 |
| False negatives | 0 | 0 |

| | | |
|---|---|---|
| True negatives | 3 | 3 |
| Accuracy | 100.0% | 100.0% |
| Precision | 100.0% | 100.0% |
| Recall | 100.0% | 100.0% |
| F1-score | 100.0% | 100.0% |
| False alarm rate | 0.0% | 0.0% |

These results indicate that all ten manually recorded classification cases were correctly classified according to the current test labels.

However, the dataset is small. The reported 100% metrics must not be interpreted as proof of 100% accuracy or reliability in real-world driving conditions.

**11. Detection-Timing Results**

The latest event_timing_log.csv contained three eye-closure events and four possible-yawn events.

| **Timing Metric** | **Eye Closure** | **Possible Yawning** |
|---|---|---|
| Number of timing events | 3 | 4 |
| Mean delay | 1.542 seconds | 1.007 seconds |
| Median delay | 1.508 seconds | 1.007 seconds |
| Minimum delay | 1.503 seconds | 1.007 seconds |
| Maximum delay | 1.615 seconds | 1.008 seconds |

The possible-yawn timing results are consistent with the configured one-second threshold. The eye-closure timing results are consistent with the configured 1.5-second threshold, with some additional delay in the recorded events.

These values describe the current software's timing records. Independent ground-truth timestamps would be needed to measure end-to-end detection delay more rigorously.

**12. Functional Testing**

The following tests have been reported as passed during project development:

1. Face detected with eyes open.
2. Normal blinking without a prolonged-closure warning.
3. Prolonged eye closure triggers a warning.
4. Alarm stops when the eyes reopen.
5. Sustained mouth opening triggers a possible-yawn warning.
6. Yawn counter increments once per continuous event.

7. Left and right head tilts.
8. Left and right face turns.
9. Recovery after the face leaves and returns to the frame.
10. Clean shutdown using the Q key.
11. Normal, dim, and bright lighting.
12. Lighting changes during operation.
13. Repeated eye-closure trials.
14. Repeated possible-yawn trials.
15. Glasses test after an adjustment to the eye-closure condition.

These are developer-reported test outcomes and have not been independently reproduced for this report.

**13. Limitations**

- The classification evaluation uses only five cases per category.
- Webcam quality, lighting, glare, camera angle, and facial occlusion can affect landmark detection.
- Facial characteristics differ between individuals, which may affect fixed thresholds.
- Mouth opening alone cannot confirm that a yawn has occurred.
- Excessive head movement may make facial measurements unreliable.
- The recorded timing delay is based on software event timestamps rather than independently measured physical onset.
- The system does not directly detect or diagnose dizziness.
- The prototype has not been validated as a safety-critical driver-monitoring system.

**14. Future Scope**

The following improvements could strengthen the project:

- Expand testing to include more participants and more positive and negative cases.
- Evaluate longer sessions under different lighting and camera positions.
- Test glasses, partial face occlusion, and a wider range of head poses.
- Record independently verified event start times for more reliable timing evaluation.
- Measure false alarms and missed detections over longer sessions.
- Add a dashboard to visualize event counts, detection delays, and test results.
- Improve calibration for different users and camera placements.
- Explore additional validated indicators of driver fatigue.

- Assess privacy, reliability, and deployment requirements before real-world use.

**15. Conclusion**

The Driver Dizziness Detection project demonstrates a real-time computer-vision prototype that uses facial landmarks to estimate eye and mouth openness, detect prolonged visual conditions, issue audible warnings, and record results for evaluation.

The current evaluation scripts successfully calculate classification metrics and summarize detection timing. The reported functional tests also indicate progress in handling moderate head movements, lighting variations, and glasses.

The next priority is to expand the test dataset, use independently verified event labels, and assess the system across more varied conditions. These steps are necessary before making strong claims about reliability in practical driving environments.

**Project status:** Working prototype with preliminary evaluation results.

**Safety note:** This prototype is an experimental monitoring aid, not a medical diagnostic tool or a replacement for safe driving practices.